



Effects of the spatial distribution of internal water storage on soil–plant hydraulic function

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HIGHLIGHTS

- SPAC model explicitly resolves water storage in root and shoot compartments.
- Internal storage is shown to strongly regulate plant hydraulic function.
- Root and shoot storage jointly buffer diurnal water-supply and demand mismatch.
- Removing storage greatly increases plant hydraulic stress under field conditions.
- Neglecting storage underestimates drought resistance and overpredicts hydraulic failure.

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ABSTRACT

Plant internal water storage buffers hydraulic disequilibrium between soil water supply and atmospheric demand, yet its contribution remains poorly quantified, especially in roots. Soil-plant-atmosphere continuum (SPAC) models commonly equate root uptake with basal sap flux, neglecting root storage and potentially biasing predictions of diurnal water fluxes and hydraulic stress. Here, we developed a coupled SPAC model that explicitly resolves water storage in root and shoot compartments. The model was evaluated under field atmospheric and soil moisture conditions to quantify the individual and combined roles of root and shoot storage in plant hydraulic function. Our results show that internal storage supplied 51% of early-day transpiration (35.5% shoot, 15.5% root), and removing either compartment reduced transpiration despite compensatory increases in the remaining storage use and root uptake. However, eliminating root and shoot storage individually increased cumulative hydraulic stress by up to 129% and 176%, respectively, relative to the full-storage baseline under critical water potential thresholds. A dry-down analysis showed that reliance on storage use increased by 37% under soil moisture limitation. These results suggest that internal storage is a key regulator of plant hydraulic function and drought stress mitigation. Models that neglect storage in roots or shoot underestimate drought resistance and overpredict hydraulic failure risk, limiting their ability to scale plant hydraulics to ecosystem processes.

1. Introduction

Plants store water within their tissues to buffer against fluctuations in water availability and transpiration demand (Hartzell et al., 2017; Kaner et al., 2020). Sap flow velocities in plants are relatively low, ranging from 0.1 to 5 m/h, which can limit the ability to sustain high transpiration rates during the day (Meinzer et al., 2004; Granier, 1987; Dye et al., 1996). Transpiration losses during day typically exceed root water uptake (Lhomme et al., 2001). During this period, plants often rely on stored water in different parts of the plants, viz., roots (Meinzer

et al., 2001; Zweifel et al., 2000), stems (Kobayashi and Tanaka, 2001), and leaves (Himeno et al., 2017) to meet their high transpiration demand. This reliance on stored water is particularly significant in large woody plants, creating a characteristic time lag between basal sap flux and canopy transpiration (Goldstein et al., 1998; Čermák et al., 2007; Köcher et al., 2013). The additional flux from storage helps plants to partially mitigate stress during daytime and sustain photosynthetic rates (Holbrook, 1995; Goldstein et al., 1998; Stratton et al., 2000; Maherali and DeLucia, 2001). During late afternoon and night when the

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transpiratory demand is less, plants uptake more water than they transpire (Running, 1979; Huang et al., 2017). A portion of the additional uptake of water termed as nocturnal or nighttime sap flow, goes into the refilling of storage in the plants. Many authors described this phenomenon in plants earlier (Waring and Running, 1978; Steinberg et al., 1990) and more recently (Phillips et al., 2003; Matheny et al., 2015).

Several approaches have been developed to quantify and model stem water storage. One such approach is the resistance-capacitance (R-C) model, which incorporates hydraulic capacitance to simulate water storage and release in stems (Carlson and Lynn, 1991; Hunt Jr et al., 1991; Manzoni et al., 2014a; Hartzell et al., 2017; Phillips et al., 2004; Gao et al., 2021). These models often reduce the spatial and temporal dynamics of tree water transport into few lumped components. As a result, they fail to accurately represent the dynamic behaviour of a tree's hydraulic storage (Chuang et al., 2006). While most R-C models focused only on aboveground storage (Carlson and Lynn, 1991; Hunt Jr et al., 1991; Manzoni et al., 2014a; Chuang et al., 2006), Gao et al. (2021) extended this approach to quantify water storage in roots. However, their model stayed within the R-C framework and still lacked physically based storage dynamics. Due to the limitations of R-C models, researchers have turned to more sophisticated models, such as porous media models treating xylem tracheids as porous media (Aumann and Ford, 2002; Bohrer et al., 2005; Manzoni et al., 2014a,b; Mirfenderesgi et al., 2016; Huang et al., 2017; Hartzell et al., 2017; Drewry et al., 2010). These models employ mass and momentum conserving partial differential equations (PDEs) to describe the spatio-temporal dynamics of water transport in plant stems. Porous media models effectively capture the lag between transpiration and sap flux while ensuring that total mass is conserved throughout the calculations. However, most of these models either solely focused on aboveground plant water storage or assumed roots to be simple conduits with no storage (Kumagai, 2001; Chuang et al., 2006; Huang et al., 2017; Vogel et al., 2017).

The contribution of plant water storage fluxes to daily transpiration varies widely among species, with aboveground storage contributing 2–50% of daily transpiration (Loustau et al., 1996; Goldstein et al., 1998;

Kobayashi and Tanaka, 2001; Holbrook and Sinclair, 1992; Maherali and DeLucia, 2001). This contribution depends on tree size (sapwood area and volume), with larger trees generally showing greater storage use (Goldstein et al., 1998). Experimental studies show that aboveground biomass and storage are strongly correlated with both tree height and D^2 (where D is the stem diameter at breast height), with D^2 typically showing stronger relationships due to its direct link to stem cross-sectional area and volume (Konôpka et al., 2011). Shoot biomass scales positively with D^2 across multiple species (Fig. 1 a). Root biomass increases strongly with aboveground biomass (Fig. 1 b), and constitutes a significant portion of the tree's total biomass.

Root biomass, on average, accounts for approximately 47% of total tree biomass depending on the biome and plant type (Qi et al., 2019). Aboveground storage contribution to daily transpiration shows a general increase with D^2 (Fig. 1 c), although substantial scatter in the data indicates considerable interspecies variability in storage strategies. The large root biomass, which scales allometrically with shoot biomass, represents a significant potential for water storage. The contribution of root water storage to daily transpiration has been challenging to quantify and is often overlooked in experimental studies, usually assuming to be small relative to that of the stems and leaves (Scholz et al., 2011). However, limited experimental evidence suggests its role is significant. Early work by Waring et al. (1979) demonstrated that roots have considerable storage capacities, accounting for 10–20% of a plant's total water storage. More recently, the experimental findings of Baer et al. (2021) suggested that roots can possess an even higher water storage capacity than the shoot.

Our review shows that while the effect of shoot storage on plant water relations is well-studied, root water storage remains poorly quantified and oversimplified in models and experimental studies. Including root storage explicitly in models is essential for predicting plant responses to climate variability. The present study aims to quantify plant water storage by explicitly modelling storage fluxes in both shoot and root compartments. It evaluates the hydraulic impacts of these storage components, with particular emphasis on the often-overlooked

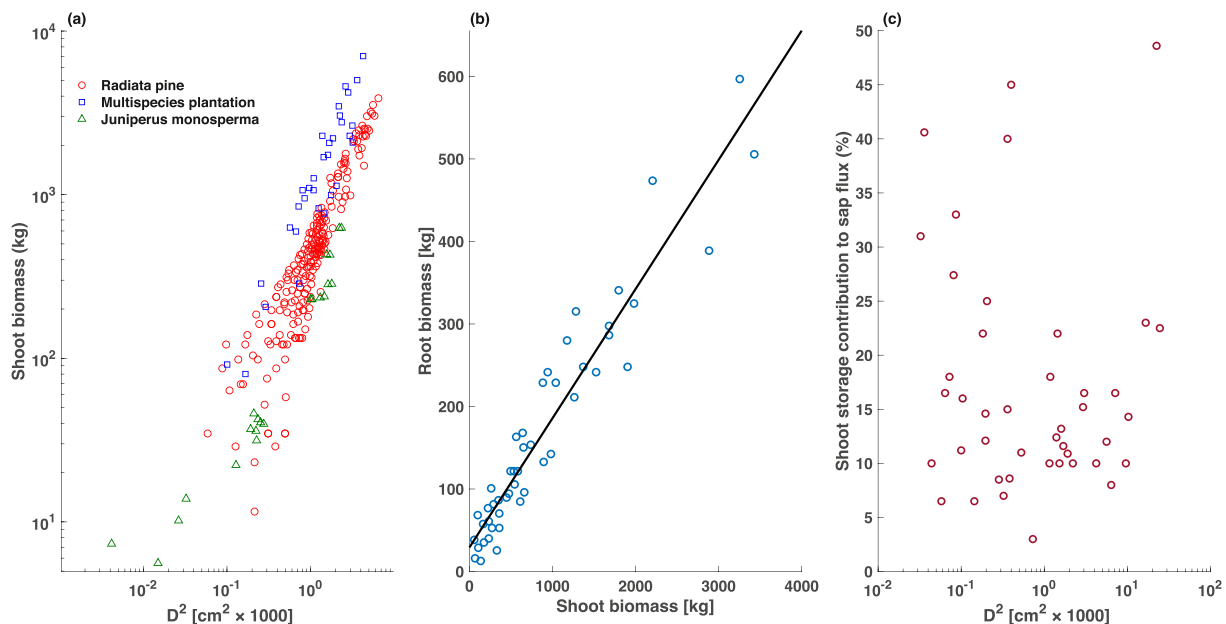


Fig. 1. Root-shoot allometric relationship to signify root water storage importance. (a) Relationship between shoot biomass and D^2 based on experimental data from multiple tree species (Moore, 2010; Mukuralinda et al., 2021; Cunliffe et al., 2020). (b) Root biomass as a function of shoot biomass, compiled from studies reporting allometric relationships across species (Qi et al., 2019; Marziliano et al., 2015). (c) Experimental data from a range of woody species to show shoot storage flux contribution to total sap flux as a function of D^2 , illustrating an overall trend with substantial interspecies variability. The data are compiled from the literature (Čermák et al., 2007; Goldstein et al., 1998; Waring and Running, 1978; Waring et al., 1979). Complete datasets for all species, including allometric relationships and storage contributions, are provided in Supplementary Dataset Files S1-S3.

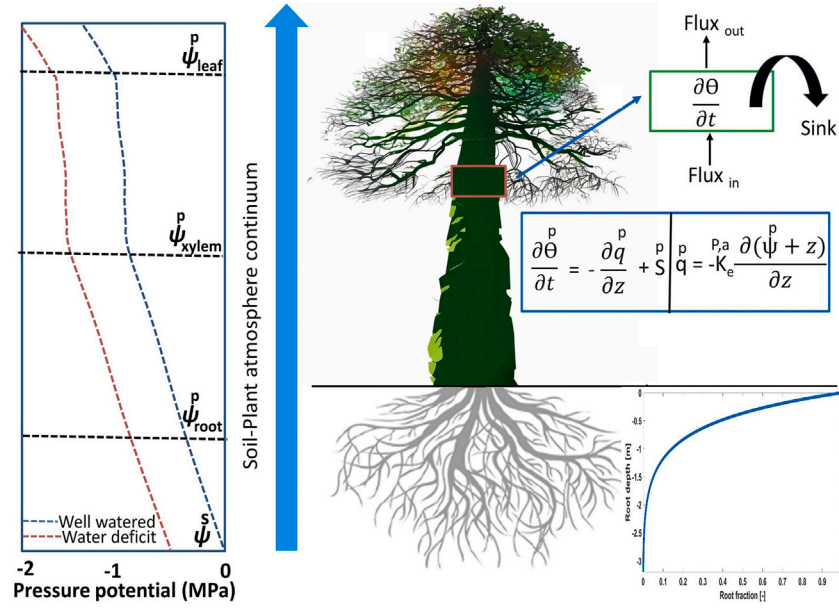


Fig. 2. Schematic representation of water movement across the soil–plant–atmosphere continuum (SPAC). Water transport is driven by gradients in water potential (ψ), with the reference potential set to atmospheric pressure (0.101 MPa). As water moves upward from the groundwater table ($\psi = 0$) through the unsaturated soil, roots, xylem, and leaves to the open atmosphere, water potential progressively decreases, becoming increasingly negative (sub-atmospheric pressure). The left panel illustrates typical water potential profiles under well-watered (blue) and water-deficit (red) conditions (adapted from Scharwies and Dinneny (2019)). The right panels show the equations for plant water dynamics, fluxes in and out of the plant, and root density distribution. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article).

root storage. We developed a one-dimensional (1D) plant hydraulic model explicitly accounting for water storage in roots and shoot. To capture the plant hydrodynamics under varying soil moisture conditions, we coupled this model with soil model using 1D Richards' equation. Additionally, to account for water loss to the atmosphere, we coupled it with a transpiration model using a vertically distributed Penman–Monteith formulation. The coupled model (vSPAC-1D) simulates the spatio-temporal (vertical) dynamics of water storage and fluxes among the soil, root, shoot, and atmosphere, explicitly accounting for root storage as an integral component of plant water dynamics.

2. Methods

A one-dimensional coupled porous media model is used to simulate water fluxes in soil and plant. It incorporates vertical variations in the hydraulic properties of soil, roots, and shoot. Governing equations are formulated using mass balance through continuity equation combined with momentum balance through Darcy's law. The model is solved using the finite difference method with mass-conserving fluxes. The equations include additional sink or source terms to account for the transfer of water between the soil and the roots, as well as between the leaves and the atmosphere. A schematic representation of the model is shown in Fig. 2, highlighting key equations and components used in the model.

2.1. Governing equations for soil moisture

The transient one-dimensional (1D) vertical flow of water in a variably saturated porous medium, with the vertical coordinate z defined as positive upward from the soil surface, is governed by Richards' equation:

$$\frac{\partial \theta}{\partial t} = \frac{\partial}{\partial z} \left[K(\psi) \left(\frac{\partial \psi}{\partial z} + 1 \right) \right] + S, \quad (1)$$

where superscript s implies soil, θ [-] denotes the volumetric soil water content, ψ [m] is the soil matric potential, K [m s⁻¹] represents

the unsaturated soil hydraulic conductivity, and S [s⁻¹] is a volumetric source/sink term accounting for water exchange with the roots. The constitutive relationships linking θ , and K to ψ are described using the van Genuchten–Mualem model (Van Genuchten, 1987; Mualem, 1976). The water retention curve is given by:

$$\theta(\psi) = \theta_r + \frac{\theta_s - \theta_r}{\left[1 + \left(\alpha |\psi| \right)^n \right]^m}, \quad (2)$$

where θ_r [-] and θ_s [-] are the residual and saturated water contents, respectively, α [m⁻¹] is a fitting parameter that is inversely proportional to the air-entry suction, n [-] characterizes the pore-size distribution, and $m = 1 - 1/n$.

The unsaturated hydraulic conductivity function, based on the Mualem pore-connectivity model (Mualem, 1976), is:

$$K(\psi) = K_{\max} \sqrt{S_e} \left[1 - \left(1 - S_e^{\frac{1}{m}} \right)^m \right]^2, \quad (3)$$

where $K_{\max}$ [m s⁻¹] is the saturated hydraulic conductivity of soil, and S_e is the effective saturation [-] defined as:

$$S_e = \frac{\theta - \theta_r}{\theta_s - \theta_r} = \left[1 + \left(\alpha |\psi| \right)^n \right]^{-m}. \quad (4)$$

Differentiating Eq. (2) yields the specific moisture capacity $C(\psi)$:

$$C(\psi) = \frac{\partial \theta}{\partial \psi} = \alpha m n \left(\theta_s - \theta_r \right) S_e^{\frac{1}{m}} \left(1 - S_e^{\frac{1}{m}} \right)^m, \quad (5)$$

Eq. (1) can be reformulated in terms of head based formulation using Eq. (5):

$$C(\psi) \frac{\partial \psi}{\partial t} = \frac{\partial}{\partial z} \left[K(\psi) \left(\frac{\partial \psi}{\partial z} + 1 \right) \right] + S. \quad (6)$$

The corresponding initial condition is given as:

$${}^s\psi(z, 0) = {}^s\psi_0(z), \quad z \in [z_{1s}, z_{ns}], \quad (7)$$

where ${}^s\psi_0(z)$ [m] represents the initial soil water potential distribution.

Boundary conditions are applied at the lower ($z = z_{1s}$) and upper ($z = z_{ns}$) limits of the domain (Fig. 4), as Dirichlet-type (prescribed potential) and Cauchy-type (prescribed flux). At the lower boundary:

$${}^s\psi(z_{1s}, t) = {}^s\psi_{z_{1s}}(t) \quad (\text{Dirichlet}), \quad (8)$$

where ${}^s\psi_{z_{1s}}$ [m] is the imposed potential.

At the upper boundary:

$$-K({}^s\psi) \left(\frac{\partial {}^s\psi}{\partial z} + 1 \right) \Big|_{z=z_{ns}} = {}^s q_{z_{ns}}(t) \quad (\text{Cauchy}), \quad (9)$$

where ${}^s q_{z_{ns}}$ [m s^{-1}] is the the imposed flux limited by the surface infiltration capacity at the infiltration-driven upper boundary (Amenu and Kumar, 2008):

$${}^s q_{z_{ns}}(t) = \min \left[r, \frac{(\theta_s - \theta(z_{ns}, t)) \Delta z}{\Delta t} \right], \quad (10)$$

where r [m s^{-1}] is the precipitation intensity, Δz [m] is the vertical grid spacing, Δt [s] is the time step, and $\theta(z_{ns}, t)$ [-] is the water content at the soil surface.

2.2. Governing equations for plant

Plant water flow is modelled with a continuity equation and Darcy's law, using the analogy between xylem sap flow and porous-media transport (Schulte and Costa, 1996; Kumagai, 2001; Aumann and Ford, 2002; Bohrer et al., 2005). Root and shoot domains are represented within a single one-dimensional vertical framework, allowing for spatially varying hydraulic properties, and source/sink terms. The transient governing equation is given as:

$$\frac{\partial \theta^p}{\partial t} = \frac{\partial}{\partial z} \left[K_e^p({}^p\psi) \left(\frac{\partial {}^p\psi}{\partial z} + 1 \right) \right] + S, \quad (11)$$

where superscript p implies plant, θ^p [-] is the volumetric water content within the plant, ${}^p\psi$ [m] is the xylem water potential (Huang et al., 2017), K_e^p [m s^{-1}] is the effective axial hydraulic conductivity, and S [s^{-1}] is the internal source/sink, capturing root water uptake (source) and transpiration losses from the canopy (sink). The effective axial conductivity K_e^p [m s^{-1}] is defined as:

$$K_e^p = K^p({}^p\psi) A_{\text{ind,cs}}^p, \quad (12)$$

where K^p [$\text{m}^3 \text{s}^{-1} / \text{m}^2_{\text{stem,cs}}$] is the specific axial conductivity of xylem tissues and $A_{\text{ind,cs}}^p$ [$\text{m}^2_{\text{stem,cs}} / \text{m}^2_{\text{ground}}$] is the stem cross-sectional area index. $A_{\text{ind,cs}}^p$ takes distinct values for the root ($z \leq 0$) and shoot ($z > 0$) segments within the unified plant domain, based on site-specific measurements of organ cross-sectional area (Macinnis-Ng et al., 2010; Yunusa et al., 2012).

The dependence of K^p on ${}^p\psi$ accounts for loss of conductivity and cavitation under stress and is described by a sigmoidal vulnerability curve (Pammenter and Van der Willigen, 1998):

$$K^p({}^p\psi) = K_{\text{max}}^p \left[1 - \frac{1}{1 + \exp[a_p({}^p\psi - b_p)]} \right], \quad (13)$$

where K_{max}^p [m s^{-1}] is the maximum specific conductivity, and a_p [m^{-1}] and b_p [m] are empirical parameters. The xylem water content (θ) is

related to ${}^p\psi$ through an empirical relationship given by (Bohrer et al., 2005):

$$\theta({}^p\psi) = A_{\text{ind,cs}}^p \theta_s^p \left(\frac{\psi_{\text{ref}} - {}^p\psi}{\psi_{\text{ref}} - \psi^p} \right)^b, \quad (14)$$

where θ_s^p [$\text{m}^3 \text{m}^{-3}$] is the saturated water content of the plant, and ψ_{ref} [m] and b [-] are empirical parameters.

Eq. (11) can be reformulated in terms of head based formulation using the plant hydraulic capacitance, which is defined as the ratio of change in xylem water content (θ) with the water potential (${}^p\psi$), i.e., $C({}^p\psi) = \partial \theta / \partial {}^p\psi$:

$$C({}^p\psi) \frac{\partial {}^p\psi}{\partial t} = \frac{\partial}{\partial z} \left[K_e^p({}^p\psi) \left(\frac{\partial {}^p\psi}{\partial z} + 1 \right) \right] + S. \quad (15)$$

Plant hydraulic capacitance quantifies elastic water storage and release in plants. To isolate the contribution of each storage compartment, (Eq. 15) reduces to: $0 = \frac{\partial}{\partial z} \left[K_e^p({}^p\psi) \left(\frac{\partial {}^p\psi}{\partial z} + 1 \right) \right] + S$, eliminating capacitive storage while preserving the full hydraulic pathway. $C({}^p\psi)$ is modelled as (Chuang et al., 2006):

$$C({}^p\psi) = A_{\text{ind,cs}}^p \frac{b \theta_s^p}{\psi_{\text{ref}} - {}^p\psi} \left(\frac{\psi_{\text{ref}} - {}^p\psi}{\psi_{\text{ref}} - \psi^p} \right)^{(b+1)}. \quad (16)$$

Since $A_{\text{ind,cs}}^p$ appears in both Eqs. (12) and (16), this single parameter simultaneously differentiates hydraulic conductance and elastic water storage capacity between root and shoot compartments within the unified framework.

The corresponding initial condition is given as:

$${}^p\psi(0, z) = {}^p\psi_0(z), \quad z \in [z_{1p}, z_{np}], \quad (17)$$

where ${}^p\psi_0(z)$ [m] denotes the initial plant water potential distribution.

Boundary conditions are applied at the lower ($z = z_{1p}$) and upper ($z = z_{np}$) limits of the domain (Fig. 4).

At the lower boundary:

$$-K_e^p({}^p\psi) \left(\frac{\partial {}^p\psi}{\partial z} + 1 \right) \Big|_{z=z_{1p}} = 0 \quad (\text{Cauchy-no flux}). \quad (18)$$

At the upper boundary:

$$-K_e^p({}^p\psi) \left(\frac{\partial {}^p\psi}{\partial z} + 1 \right) \Big|_{z=z_{np}} = 0 \quad (\text{Cauchy-no flux}). \quad (19)$$

2.3. Soil-root water uptake and exchange

Soil and plant roots are connected to each other with the source/sink term that allows water movement from soil to roots and from the roots to the soil based on the water potential difference between the two, considering equal hydraulic conductivity of the root membrane in both directions. The water flux between soil and roots depends on root radial conductivity, which represents the radial resistance to water flow between the soil and roots. The water flux between the soil and the roots can be given as:

$$S = K_e^r({}^r\psi - {}^s\psi), \quad (20)$$

where ${}^r\psi$ [m] is the root water potential and K_e^r [m s^{-1}] is the effective root radial conductivity. S [s^{-1}] can be positive or negative depending

upon its direction. At any given time, the source/sink term in the soil and root is always equal and opposite at $Z_r \leq z \leq 0$ (Fig. 4),

$$\frac{\partial}{\partial z} S = -S. \quad (21)$$

The effective root radial conductivity $K_e^{p,r}$ in any soil layer is a function of total root radial conductivity, root fraction present in that layer, and the soil moisture content (Amenu and Kumar, 2008; Verma et al., 2014; Quijano and Kumar, 2015),

$$K_e^{p,r} = K^{p,r} f(\theta) f_r(z), \quad (22)$$

where $K^{p,r}$ [$\text{m}^3 \text{s}^{-1} \text{m}^{-1} / \text{m}_{\text{root,sa}}^2$] is the total root radial conductivity, $f_r(z)$ [$\text{m}^2 \text{m}^{-3}$] is the root distribution function to account for root fraction in different layers of the soil, and $f(\theta)$ [-] is the water stress reduction function that accounts for reduction in root water uptake due to soil moisture limitation. $f(\theta)$ can be expressed as (Feddes et al., 1976):

$$f(\theta) = \begin{cases} 0, & \theta \leq \theta_w, \\ \frac{\theta - \theta_w}{\theta_f - \theta_w}, & \theta_w < \theta \leq \theta_f, \\ 1, & \theta > \theta_f, \end{cases} \quad (23)$$

where θ_w [-] is the wilting point, which represents soil water content below which root water uptake ceases, and θ_f [-] is the field capacity of the soil. $f_r(z)$ is modelled based on (Vrugt et al. (2001) one-dimensional root water uptake model.

$$f_r(z) = \frac{\beta_z}{Z_r \int_0^{\beta_z} \beta(z) dz}, \quad (24)$$

where $\beta(z)$ [-] is a shape factor that characterizes how roots are distributed spatially with depth. It can be described as (Vrugt et al., 2001),

$$\beta(z) = \left(1 - \frac{|z|}{Z_r}\right) \exp\left(\frac{-\rho_z}{Z_r} |z|\right), \quad (25)$$

where ρ_z [-] is an empirical parameter.

2.4. Canopy transpiration modelling

Water movement in aboveground xylem is modelled using a sink term representing transpiration losses from the aboveground plant. The total transpiration is distributed vertically throughout the canopy based on the leaf area density (LAD) distribution [$\text{m}^2 \text{m}^{-2} \text{m}^{-1}$] (leaf area per unit ground area per unit height), which integrates to the total canopy leaf area index (LAI), i.e.,

$$\int_0^H \text{LAD}(z) dz = \text{LAI}. \quad (26)$$

The vertically distributed water loss in the aboveground domain at $0 < z \leq H$ (Fig. 4) is given as (Chuang et al., 2006; Manoli et al., 2014):

$$\frac{\partial}{\partial z} S(z, t) = \text{LAD}(z) E(t), \quad (27)$$

where $\frac{\partial}{\partial z} S(z, t)$ is the aboveground sink term [$\text{m}^3 \text{m}^{-3} \text{s}^{-1}$], and $E(t)$ [$\text{m} \text{s}^{-1}$] is the transpiration flux. The empirical LAD(z) function is expressed as (Lalic and Mihailovic, 2004):

$$\text{LAD}(z) = l_{\max} \left(\frac{H - z_{\max}}{H - z} \right)^{n_0} \exp \left[n_0 \left(1 - \frac{H - z_{\max}}{H - z} \right) \right], \quad (28)$$

where $l_{\max}$ [$\text{m}^2 \text{m}^{-3}$] is the maximum leaf area density, $z_{\max}$ [m] is the height of maximum leaf area density, and n_0 [-] is an empirical shape

parameter:

$$n_0(z) = \begin{cases} 6, & 0 \leq z < z_{\max}, \\ 0.5, & z_{\max} \leq z \leq H. \end{cases} \quad (29)$$

Daytime transpiration is calculated using the Penman–Monteith formulation (Allen et al., 1998) coupled with the Jarvis (1976) stomatal conductance model, with daytime defined as periods when incoming solar radiation exceeds $5 \text{ [W m}^{-2}\text{]}$ (Wang et al., 2012). Nighttime transpiration is modelled with a simplified function of temperature, vapour pressure deficit, and plant water potential (see Supporting Information section S4).

2.5. Study site and data

The study site is located in New South Wales, Australia ($33^\circ 39' \text{ S}$, $150^\circ 46' \text{ E}$), dominated by *Eucalyptus parramattensis* and *Angophora bakeri*. The overstorey averages 14 [m] in height, with a leaf area index (LAI) of $1.3\text{--}1.9$. The climate is temperate with mean daily temperatures of $10\text{--}24^\circ \text{C}$ and an annual rainfall of approximately 730 [mm] . Meteorological forcing (rainfall, radiation, temperature, vapour pressure deficit) and sap flux measurements were available for January–June 2007.

All model parameters were obtained from published literature and site-specific measurements at the study site (Macinnis-Ng et al., 2010, 2011; Yunusa et al., 2012; Zeppel et al., 2008). No parameters were calibrated against observed data. Model performance was evaluated through independent validation against diurnal sap flux dynamics and nocturnal water partitioning reported for this site (Zeppel et al., 2010). Additional details on site characteristics, measurement protocols, and parameter values and sources are provided in the Supporting Information (Section S2).

2.6. Numerical implementation

The general, modular one-dimensional soil–plant–atmosphere hydraulic framework described in 2.1–2.4 was implemented to simulate vertical water transport in the soil, root, and aboveground xylem. The framework accepts different soil constitutive laws (e.g., Van Genuchten (1987); Brooks and Corey (1964)), root and leaf area density distributions, species-specific hydraulic parameters, and multiple representations of canopy demand (prescribed transpiration coupled with stomatal models such as Jarvis (1976)/Ball et al. (1987)/Leuning (1995) or other empirical formulations). Boundary conditions are configurable per case and include prescribed potential (Dirichlet), prescribed flux (Cauchy), or free-drainage (unit-gradient) at the lower boundary. In this study, we applied the model to a single site for evaluation by comparing simulated fluxes with previously published measurements. However the implementation is intentionally generic and can be applied to other species, soil types, and meteorological forcings without requiring structural changes to the code. The process flow of the model is presented in Fig. 3.

To determine the contribution of internal water storage compartments to whole plant water dynamics, we performed three numerical experiments with different storage configurations. The baseline configuration retained full capacitance in both root and shoot domains and served as a reference for transpiration buffering under natural atmospheric variability. Subsequent simulations systematically removed capacitance to isolate the functional roles of root and shoot storage and to test the plant's sensitivity to diminishing storage capacity. $\bar{C}(\psi)$ was set to zero on the respective root ($z \leq 0$) or shoot ($z > 0$) domains, as described in Section 2.2, ensuring that differences in outputs are attributable exclusively to organ-specific capacitive storage.

The system of equations used to model water flow in soil and plants is challenging to solve due to its non-linear nature, and it is generally necessary to use numerical techniques, such as finite differences

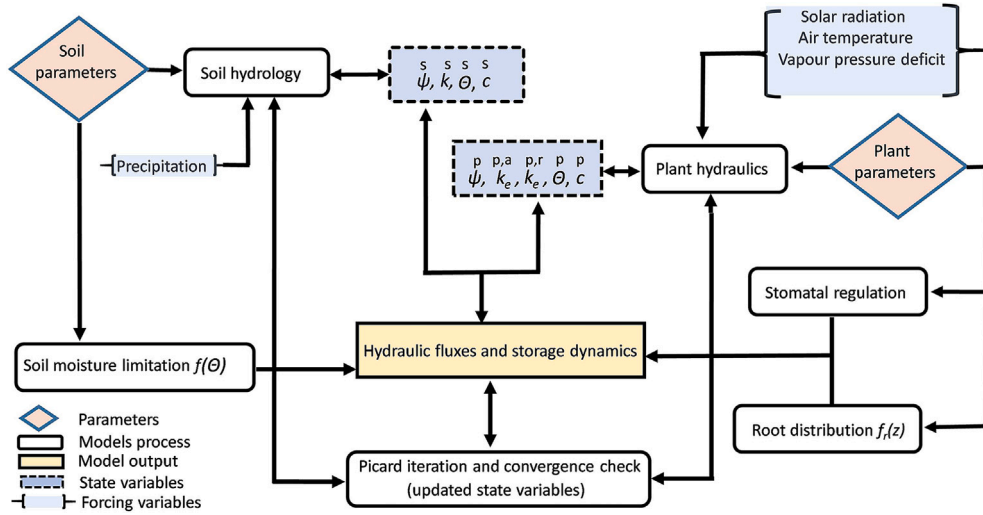


Fig. 3. Flowchart of the coupled soil-plant-transpiration model. Soil water dynamics and plant water transport are coupled through belowground soil–plant interactions Governing hydraulic fluxes and storage dynamics. Root distribution and stomatal regulation control the partitioning of fluxes, while atmospheric forcing drives transpiration.

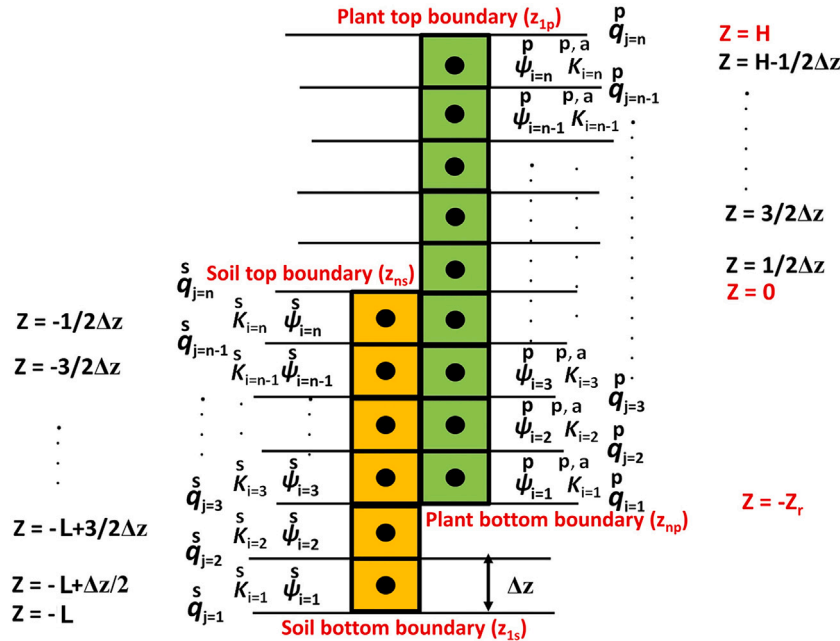


Fig. 4. Numerical discretization of the soil-plant hydraulic system. Water potentials (ψ) and hydraulic conductivities (K) are computed at nodes (indexed by i), whereas fluxes (q) are computed at interfaces (indexed by j). The domain extends from soil depth L to plant height H , with the soil-root interface located at $Z = 0$.

or finite elements, to obtain approximate solutions (Allen and Murphy, 1985). We solved these equations using the fixed step iterative method (FIM) with the first order backward Euler implicit method. The fixed step iteration employed a modified Picard iteration (Zarba, 1988) to solve the mixed form of Richards' equation (Celia et al., 1990), as detailed in the Supporting Information (section S1). The soil-plant system was divided into distinct hydraulic compartments represented by finite-difference computational cells or grids (Fig. 4). A cell-centred scheme was used, storing all state variables at grid nodes and fluxes at grid interfaces. The governing equations were solved on a uniform mesh with a spatial resolution of $\Delta z = 0.1$ [m], using a time step that ensured numerical stability and consistent flux continuity between soil, root, and shoot compartments.

3. Results

3.1. Numerical convergence and resolution analysis

A systematic grid convergence analysis was performed to identify the minimum spatial and temporal resolutions needed for numerically stable and physiologically consistent simulations in the vSPAC-1D model. Five resolution combinations were tested, with the time step Δt spanning 225–3600 [s] and the spatial discretization Δz ranging from 0.0125–0.20 [m]. Convergence was evaluated using two complementary diagnostics: (i) the vertical water potential profile, and (ii) the cumulative sap flux. Together, these metrics capture both local hydraulic gradients and whole-plant fluxes, providing a quantitative basis for determining asymptotic convergence under grid refinement.

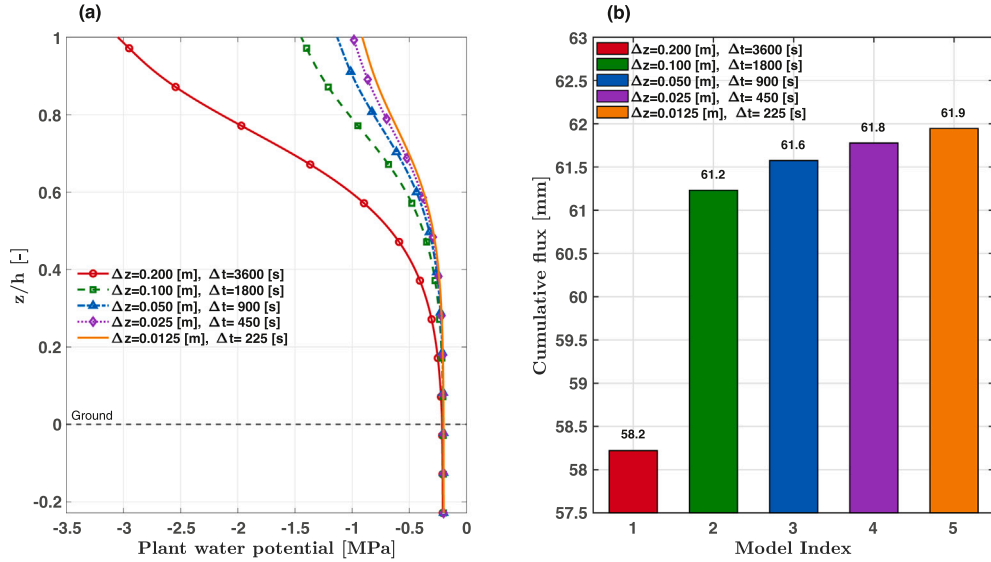


Fig. 5. Numerical convergence and resolution analysis for the plant hydraulic model. (a) Plant water potential (ψ) profiles along the normalized vertical coordinate (z/h) for five grid resolutions, showing convergence as spatial and temporal resolutions are refined. (b) Cumulative water flux, showing progressive convergence with diminishing improvements at the finest tested grid. Model indices correspond to the tested grids, from coarse (1) to fine (5) resolutions.

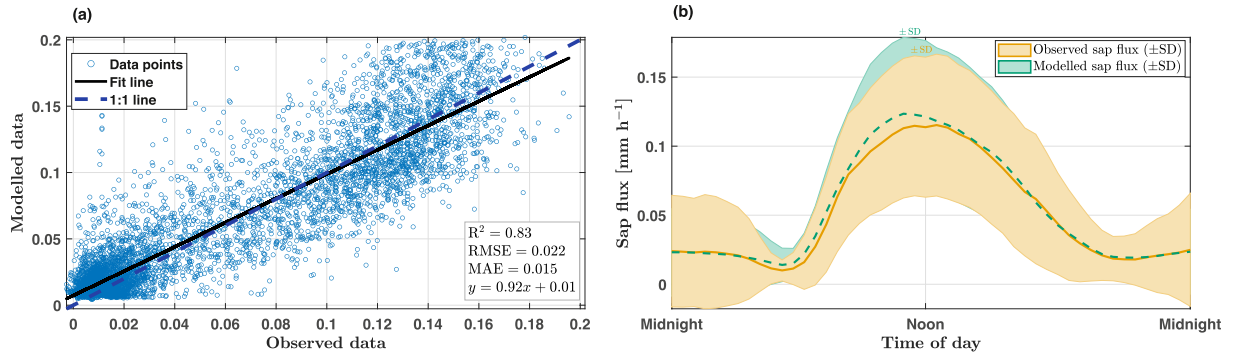


Fig. 6. Model evaluation against observed sap flux data. (a) Half-hourly comparison between observed and modelled sap flux over the full study period. The solid black line denotes the least-squares regression fit, and the dashed blue line represents the 1:1 reference line. (b) Mean diurnal cycle of sap flux over the study period. The solid line represents the observed sap flux, whereas dashed line represents modelled sap flux. The corresponding shaded regions indicate ± 1 standard deviation (SD). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article).

The distribution of plant water potential (ψ) along the normalized vertical coordinate (z/h) shows resolution-dependent differences across the five tested grids (Fig. 5 a). The coarsest discretization ($\Delta z = 0.20$ [m], $\Delta t = 3600$ [s]) deviates markedly from the finer solutions, especially in the upper portion of the shoot ($z/h > 0.4$), where the profile becomes noticeably flatter. This indicates that the coarsest grid fails to resolve the full curvature of the axial water-potential gradient, which is most sensitive to refinement in this upper segment. Intermediate resolutions progressively reduce this discrepancy, with the largest improvement occurring between the first and second tested grids.

As both spatial and temporal discretizations are refined, the profiles progressively collapse toward a single curve, demonstrating convergent behaviour consistent with a stable numerical scheme. Once the grid spacing reaches $\Delta z \leq 0.10$ [m] and the time step satisfies $\Delta t < 1800$ [s], the difference between successive solutions becomes visually indistinguishable across the entire height domain, including the region where deviations were previously most pronounced. Further refinement to the finest tested configuration ($\Delta z = 0.0125$ [m], $\Delta t = 225$ [s]) produces no additional structural change in the profile, indicating that the solution has entered the asymptotic convergence regime. At this stage,

residual differences between curves fall below the practical resolution required for the intended modelling purposes, confirming that additional refinement provides no meaningful improvement in predictive accuracy.

The coarsest configuration yields the lowest predicted flux (58.2 [mm]), showing that insufficient spatial and temporal resolution leads to a systematic underestimation of water transport through the system (Fig. 5 b). As the grid is refined, the cumulative flux increases and progressively approaches a stable reference value of approximately 62.0 [mm], indicating monotonic convergence. The most significant improvement occurs when refining from $\Delta z = 0.20$ [m] to 0.10 [m] and $\Delta t = 3600$ [s] to 1800 [s], yielding an increase of 3.0 [mm]. This confirms that the majority of the truncation error is removed at this intermediate resolution level.

Further refinement of both space ($\Delta z = 0.10$ [m] $\rightarrow$ 0.0125 [m]) and time ($\Delta t = 1800$ [s] $\rightarrow$ 225 [s]) produces only a marginal change of about 0.8 [mm], corresponding to less than 1.5% variation relative to the converged value. This behaviour indicates that the numerical solution has entered the asymptotic regime, where additional refinement yields negligible improvements in predictive accuracy relative to the

computational cost. The grid resolution $\Delta z = 0.10$ [m] combined with $\Delta t = 900$ [s] therefore provides an optimal balance, lying within the converged region while remaining within 1% of the fully refined solution. This configuration was adopted for all subsequent model simulations.

3.2. Model evaluation against field observations

The model was evaluated against observed sap flux measurements collected at the study site. The model simulations reproduced both the magnitude and temporal variability of sap flux with high accuracy (Fig. 6). The one-to-one comparison between half-hourly observed and modelled sap flux (Fig. 6 a) demonstrates a strong linear correspondence, with an R^2 of 0.83 and relatively low scatter across the full dynamic range (0–0.20 [mm h⁻¹]). The regression slope of 0.92 indicates a small but systematic underestimation of higher flux, while the positive intercept of 0.01 [mm h⁻¹] suggests slight overprediction during low-flux periods, typically associated with night-time or weak evaporative demand. Model errors appear approximately homoscedastic across the range of sap flux, with no evidence of disproportionate divergence at either low or high transpiration rates. This indicates that the model response scales consistently across varying soil-moisture and environmental conditions.

The mean diurnal sap flux pattern further confirms the model's ability to reproduce the characteristic daily behaviour of plant water use (Fig. 6 b). Both observations and simulations exhibit negligible nocturnal flux, followed by a rapid post-sunrise increase driven by stomatal opening and rising atmospheric demand. Midday fluxes are closely aligned, with observed and modelled peaks of 0.118 and 0.125 [mm h⁻¹] respectively, corresponding to a 5.9% difference at maximum rates. The afternoon decline is similarly captured, reflecting the joint influence of decreasing radiation, declining vapour pressure deficit, and progressive stomatal regulation. The shaded region (± 1 SD) shows that not only mean behaviour but also day-to-day variability is reproduced to first order, with only modest divergence during the early morning and late afternoon transition periods.

While the overall agreement is strong, the residual differences provide insights into the possible dominant mechanisms behind remaining discrepancies. The model response varies across atmospheric and physiological conditions, and part of the observed mismatch may arise from how these transitions are represented. During periods of high evaporative demand, the model relies on a simplified stomatal formulation that may not fully account for short-term hydraulic or biochemical regulation at the leaf level. Conversely, under low atmospheric demand, particularly during nighttime hours, transpiration is represented using an empirical formulation that incorporates temperature, vapour pressure deficit, and leaf water potential controls. While this formulation accounts for first-order environmental and hydraulic limitations, it does not explicitly resolve stomatal conductance as a dynamic variable at night. As a result, certain leaf-level regulatory processes, such as incomplete stomatal closure or endogenous controls, are not explicitly resolved but are implicitly captured within this formulation. These simplifications are adopted to maintain model parsimony and computational efficiency, particularly given the relatively low magnitude of nighttime fluxes, but may contribute to some of the scatter observed in these results. The magnitude of the residual errors is small in absolute terms, with an RMSE of 0.022 [mm h⁻¹] and an MAE of 0.015 [mm h⁻¹].

3.3. Partitioning of plant water fluxes

The diurnal dynamics of plant water fluxes reflect the interactions between root water uptake, internal water storage, and transpiration demand. Our results show that internal storage contributes substantially to meeting transpiration demand. This contribution is most evident during early-day transpiration periods when atmospheric demand exceeds root water uptake, creating a lag between water loss and water supply. A similar pattern has been reported previously at the same woodland site by

Zeppel et al. (2010), who found that internal storage can supply a substantial portion of daily transpiration when root uptake is temporarily insufficient during this lag period.

The temporal partitioning of water fluxes shows distinct phases over the diurnal cycle. During the morning phase, as atmospheric demand rapidly increases and root water uptake lags, internal water storage becomes the primary contributor to transpiration as simulated by the model. At this stage, model-inferred storage accounts for nearly half of the total transpiration flux, with both root and shoot compartments contributing substantially. The greater contribution from shoot storage (35.6%) relative to root storage (15.4%) indicates a functional asymmetry in hydraulic capacitance, consistent with the greater capacitance of aboveground tissues and their faster response to changes in leaf water potential.

During the midday phase, transpiration peaks at 0.15 [mm h⁻¹], and root uptake supplies most of the flux ($72.2\% \pm 4.29\%$, mean $\pm$ SE). However, the results indicate that internal storage continues to buffer the system, providing the remaining $27.8\% \pm 3\%$ of the transpiration stream (10% from root storage and 17.8% from shoot storage). This model-inferred sustained storage contribution during peak demand, particularly from root compartments, is consistent with emerging evidence that root tissues serve not only as conduits but also as capacitive tissues (Scholz et al., 2007, 2008; Meinzer et al., 2004). The model inferred storage contribution during peak atmospheric demand suggests a moderating effect on midday xylem tension, which may help keep water potentials above thresholds associated with hydraulic failure risk.

After the midday maximum, transpiration begins to decline, and the balance between supply pathways shifts accordingly. Root uptake progressively assumes the dominant role, eventually accounting for approximately 92% of the total flux, while the contribution from internal storage diminishes. This shift likely reflects two concurrent processes: the gradual depletion of the portion of stored water that is most readily available for release, and the restoration of closer hydraulic coupling between root uptake and transpiration as evaporative demand decreases. The progressive reduction in storage contribution during this period highlights the finite nature of internal reserves, reinforcing that storage is a dynamic pool that must be refilled rather than a continuously available buffer.

In the afternoon, transpiration declines and eventually drops below root water uptake, creating a positive water balance in which supply exceeds atmospheric demand. The surplus water is directed toward depleted storage compartments, with refilling becoming evident. Maximum refilling rates of 0.03–0.05 [mm h⁻¹] occur during late evening when the difference between root water uptake and transpiration is maximum. Mean nocturnal transpiration accounts for about two-thirds ($65\% \pm 4\%$) of nighttime root uptake, while the remainder is partitioned between shoot (22%) and root (13%) storage. The close agreement between modelled nocturnal transpiration (65% of root uptake, 95% CI: 56–74%) and field observations at the same site (50–70%; Zeppel et al. (2010)) validates our hydraulic framework and suggests that nocturnal processes are integral to daily water balance rather than negligible losses. Sensitivity analysis confirmed that both the storage contribution and nocturnal fraction are robust to $\pm 20\%$ uncertainty in the four key parameter groups, varying by no more than $\pm 7\%$ and $\pm 9\%$ respectively across all perturbations (Supporting Information Table S4–S5, Supporting Information Fig. S3).

3.4. Coordination among transpiration, root water uptake, and internal storage under contrasting storage configurations

Storage is selectively removed from root and shoot compartments across three scenarios: (i) both root and shoot storage enabled, (ii) no root storage, and (iii) no shoot storage. These scenarios isolate the functional role of internal storage in diurnal flux dynamics. The analysis quantifies how these changes propagate through phase relationships, peak fluxes, and the redistribution of storage use between root and

shoot compartments. Internal water storage is a key regulator of the diurnal water balance of plants, buffering the mismatch between transpiration demand and soil-root water supply. When storage is available, the plant can transiently decouple water loss from uptake, allowing it to sustain higher transpiration rates than would be possible through instantaneous root uptake alone. In the simulated removal scenarios, this hydraulic autonomy diminishes as storage is reduced, with fluxes becoming increasingly constrained by real time soil supply, peak transpiration declining, and the temporal separation between uptake and loss narrowing.

A defining feature across all scenarios is the clear phase lag between transpiration and RWU, with transpiration consistently preceding RWU by approximately 0.5–3 [h]. This temporal decoupling is most evident in the complete storage scenario (panel a), where peak transpiration reaches $0.15 \text{ [mm h}^{-1}\text{]}$, while RWU peaks after 3 [h] with a marginally lower maximum flux ($0.12 \text{ [mm h}^{-1}\text{]}$). The systematic reduction in phase lag from 3 [h] (with both storages) to 2.5 [h] (without root storage) to 0.5 [h] (without shoot storage) suggests that shoot storage acts as the primary temporal buffer between atmospheric demand and soil water supply, while root storage provides supplementary temporal buffering. The persistence of the lag confirms that uptake is limited by soil-plant transport, whereas transpiration is driven directly by atmospheric forcing, and storage mediates the mismatch between the two.

When root storage is disabled (panel b), peak transpiration declines from 0.15 to $0.14 \text{ [mm h}^{-1}\text{]}$, a reduction of about 7%, despite compensatory increases in both shoot storage flux (15%) and RWU (7.5%). This partial compensation is insufficient to recover the original flux magnitude, indicating that root storage, although smaller than shoot storage in absolute terms, still plays a measurable role in sustaining daytime water loss in this framework. The inability of shoot storage to fully compensate suggests that the spatial distribution of storage capacity along the hydraulic pathway is functionally important, not just the total capacitance. Panel (e) shows that when root storage is disabled, the total plant storage flux naturally becomes identical to shoot storage flux

alone, reflecting complete reliance on aboveground storage to absorb the RWU-transpiration imbalance.

When shoot storage is disabled (panel c), the changes in diurnal fluxes become more pronounced. Peak transpiration decreases to $0.13 \text{ [mm h}^{-1}\text{]}$, while peak RWU increases from 0.12 to $0.13 \text{ [mm h}^{-1}\text{]}$. The phase lag between the two fluxes collapses from 3 to <1 [h], resulting in much closer coupling between them. This pattern reflects the dominant buffering role of shoot storage. When transpiration increases, shoot storage releases water to meet immediate demand while allowing RWU to increase more gradually. Without this buffer, roots must respond almost immediately to changes in transpiration, explaining both the increased RWU peak and the synchronized flux patterns. Once shoot storage is removed, the total plant storage flux becomes identical to the root storage flux (panel f), as expected when the shoot contribution is zero. Root storage flux increases by 50%, representing a substantial relative increase in an attempt to compensate for the missing shoot storage, but with limited success. This suggests that although root storage can partially offset the loss of shoot storage, its limited capacity in this framework cannot replicate the buffering function typically provided by shoot storage.

The differences revealed by the storage configurations (Fig. 8) highlight key hydraulic mechanisms relevant to plant performance under water-limited conditions. The systematic decrease in transpiration (from 0.15 to $0.13 \text{ [mm h}^{-1}\text{]}$), accompanied by an increase in RWU (from 0.12 to $0.13 \text{ [mm h}^{-1}\text{]}$) as storage components are removed, shows that distributed capacitance enables plants to sustain higher transpiration rates than would be possible through instantaneous soil-root water transport alone. The compensation patterns also clarify the distinct roles of the two storage compartments: shoot storage provides the dominant temporal buffering, maintaining a 2.5–3 [h] phase lag and supporting higher absolute fluxes, while root storage contributes smaller fluxes but still exerts a measurable influence on the soil-plant hydraulic balance. Even with compensatory increases in shoot storage flux and RWU when root storage is removed, transpiration declines by 7%, suggesting that other

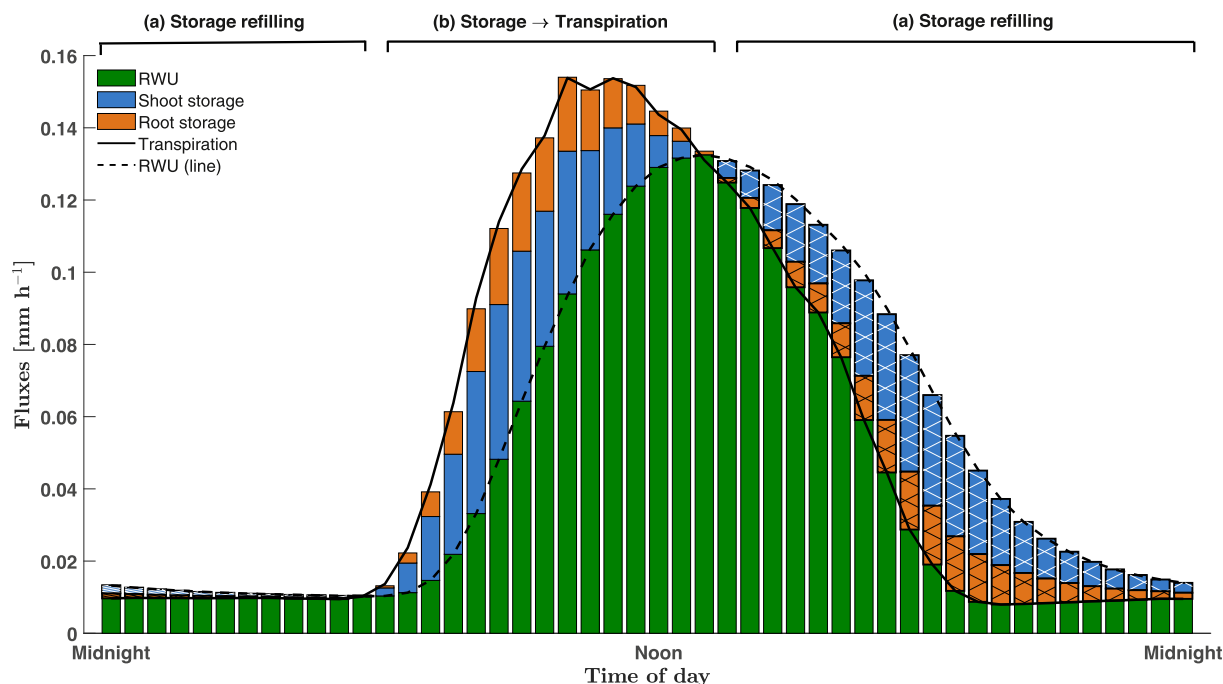


Fig. 7. Diurnal dynamics of plant water fluxes. Stacked bars represent the half-hourly contributions of root water uptake (green), shoot storage flux (blue), and root storage flux (Orange) to meet transpiration demand. The solid black line denotes the transpiration rate, while the dashed black line shows total root water uptake. (a) Cross-hatched segments indicate periods when uptake exceeds transpiration, allowing refilling of internal water storage. (b) Solid stacked bars (morning to midday) indicate when internal storage supplements root uptake to meet transpiration demand. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article).

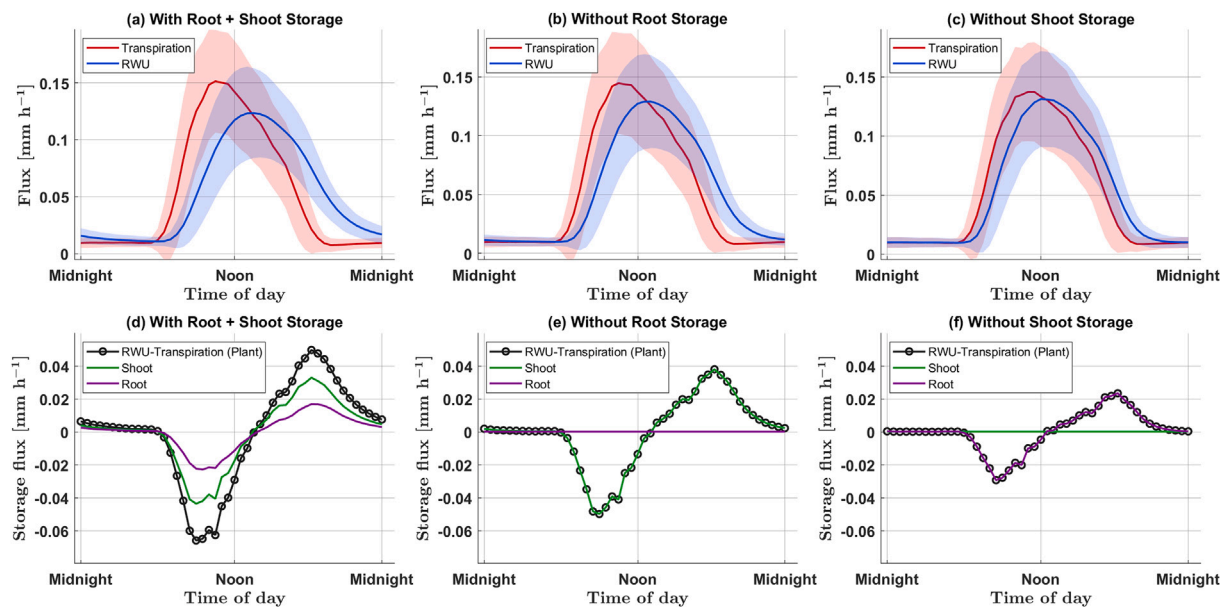


Fig. 8. Mean diurnal cycles of transpiration and root water uptake (RWU) over the study period. Shaded regions show ± 1 standard deviation across days. The three columns compare storage configurations: (a, d) both root and shoot storage enabled, (b, e) no root storage, and (c, f) no shoot storage. The top row (a–c) shows transpiration (red) and RWU (blue). The bottom row (d–f) shows the corresponding storage fluxes, partitioned into plant (black), shoot (green), and root (purple) components. Storage buffers daytime water deficits when transpiration exceeds RWU and refills when RWU exceeds transpiration, primarily during the afternoon through night. Positive storage flux indicates refilling, while negative values indicate release. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article).

hydraulic components cannot fully compensate for the root contribution in this framework. The 50% increase in root storage flux when operating alone suggests that root storage normally operates well below capacity and may serve as a hydraulic reserve during periods of high evaporative demand.

3.5. Dynamic role of internal water storage in regulating plant water potential

The rate of change of leaf water potential ($d\psi/dt$) diverged strongly among storage configurations (Fig. 9 a). During the morning stress development phase, plants without shoot storage showed the steepest decline (-0.60 MPa h^{-1}), those without root storage declined at an intermediate rate (-0.50 MPa h^{-1}), and plants with full storage declined most slowly (-0.44 MPa h^{-1}). These differences show that internal water storage slows the rate at which water potential responds to increasing transpirational demand, with shoot storage providing more immediate buffering than root storage. During the recovery phase, the pattern reversed: plants without storage recovered faster than those with full storage. This occurs because capacitive tissues must themselves be refilled during recovery, effectively spreading the recovery process over a longer time period. This reflects a fundamental trade-off in plant hydraulics: storage slows both stress development and recovery.

Quantification of peak rates confirmed the magnitude of these dynamic effects (Fig. 9(b)). Shoot storage removal increased maximum stress development rates by 38% compared to full storage (from 0.44 to $0.60 \text{ [MPa h}^{-1}\text{]}$), while root storage removal increased them by 13% (to $0.495 \text{ [MPa h}^{-1}\text{]}$). These rate differences are physiologically significant because a 38% faster rate of stress development means plants reach critical hydraulic thresholds substantially sooner, potentially before protective mechanisms such as stomatal closure can fully engage. Recovery rates showed similar magnitudes of increase when storage was removed (35% and 13% for shoot and root storage removal, respectively), confirming that storage dampens rates of change in both directions.

The most consequential effect of storage emerged in the analysis of cumulative hydraulic stress (Fig. 9 c). We quantified cumulative stress as the time integrated magnitude of water potential at three ecologically grounded thresholds. The primary threshold of $\psi_{50} = -2.0 \text{ [MPa]}$ represents the estimated 50% hydraulic conductivity loss for the dominant woodland species at this site (Choat et al., 2012; Zhang et al., 2016). Two additional thresholds were selected to assess robustness: -1.8 [MPa] , representing a point of significant hydraulic conductivity decline in *Eucalyptus* species (Zhang et al., 2016), and -1.9 [MPa] , corresponding to the measured minimum leaf water potential of *Angophora bakeri* at this site (Zeppel et al., 2010). At -1.8 [MPa] , root and shoot storage removal increased model-inferred cumulative stress by 25% and 34% respectively, and at -1.9 [MPa] by 43% and 60%. The sharp escalation at $\psi_{50} = -2.0 \text{ [MPa]}$, where root and shoot storage removal increased cumulative stress by 129% and 176% respectively, reflects the fact that spatially distributed storage keeps the plant operating just above this critical threshold in this framework. Full storage barely breached this threshold while storage removal pushed the plant substantially beyond it, disproportionately amplifying cumulative exposure at this level of hydraulic stress. Across all three thresholds, storage consistently reduced cumulative hydraulic stress, with the difference between scenarios growing larger at more negative water potentials. In this framework, these model-inferred increases indicate that neglecting shoot storage in the model leads to nearly a 2.8 times increase in cumulative exposure to hydraulically damaging water potentials, while omission of root storage results in more than a twofold increase. Sensitivity analysis confirmed that the relative stress differences between scenarios remain robust to parameter uncertainty (Supporting Information Tables S4–S5, Supporting Information Fig. S3), supporting the robustness of inferred storage effects on hydraulic stress across the evaluated model configurations.

These results underscore not only the dominant role of shoot storage, but also the substantial contribution of root storage to hydraulic safety within the modelling framework. Because embolism formation depends on both the magnitude and the duration of xylem tension, these increases

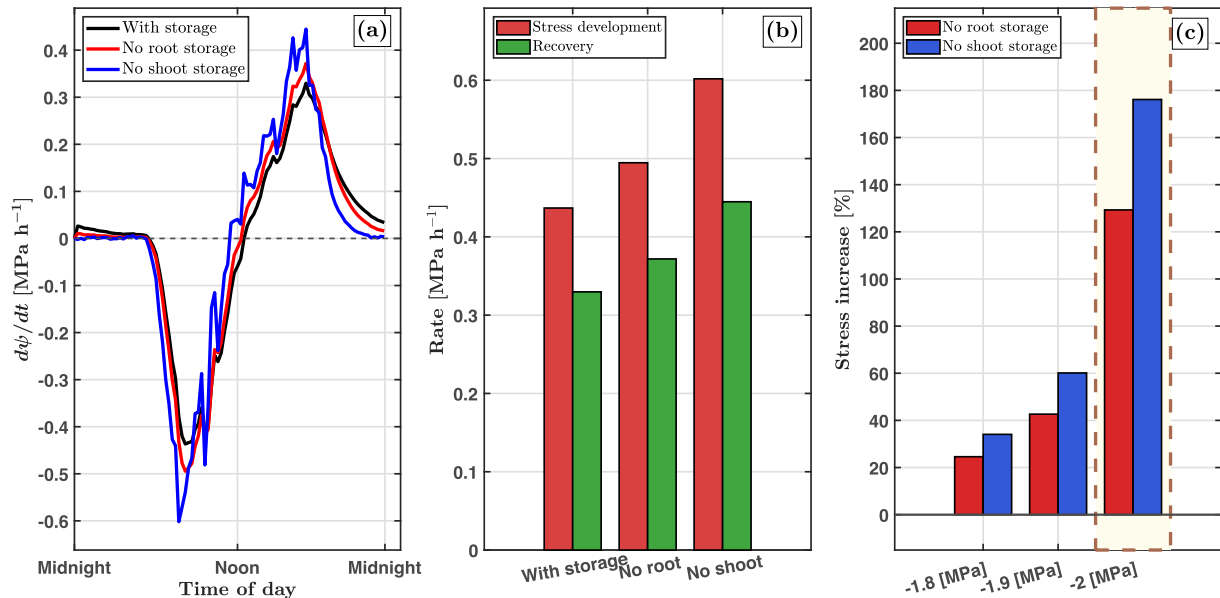


Fig. 9. Dynamic effects of internal storage on water potential regulation. (a) Diurnal rate of change of leaf water potential ($d\psi/dt$) for full storage, no root storage, and no shoot storage scenarios. (b) Peak rates of stress development (red) and recovery (green), highlighting the buffering effect of storage on hydraulic fluctuations. (c) Relative increase in cumulative hydraulic stress at three water potential thresholds (-1.8 , -1.9 , and $\psi_{50} = -2.0$ [MPa]), expressed as percentage increase relative to the full storage baseline. The shaded region highlights the ψ_{50} threshold. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article).

in cumulative stress indicate a substantial escalation in hydraulic failure risk in this framework. Under prolonged drought, where severe water tension persists across successive days, these effects may compound, suggesting that failure to represent internal storage processes could lead to biased predictions of plant hydraulic response. This interpretation is consistent with experimental studies demonstrating that plant water storage acts as a hydraulic capacitor, buffering the drop in water potential during periods of high transpirational demand and thereby reducing exposure to damaging low potentials and delaying hydraulic

failure under high evaporative demand (Tyree and Ewers, 1991; Meinzer et al., 2004; Scholz et al., 2007).

3.6. Response of plant water fluxes to soil moisture limitation

Soil moisture limitation exerted a strong control on water fluxes within the soil-plant continuum (Fig. 10). A progressive reduction in soil water availability induced a coordinated response of hydraulic fluxes, linking soil moisture decline to shifts in root uptake and increased use of internal

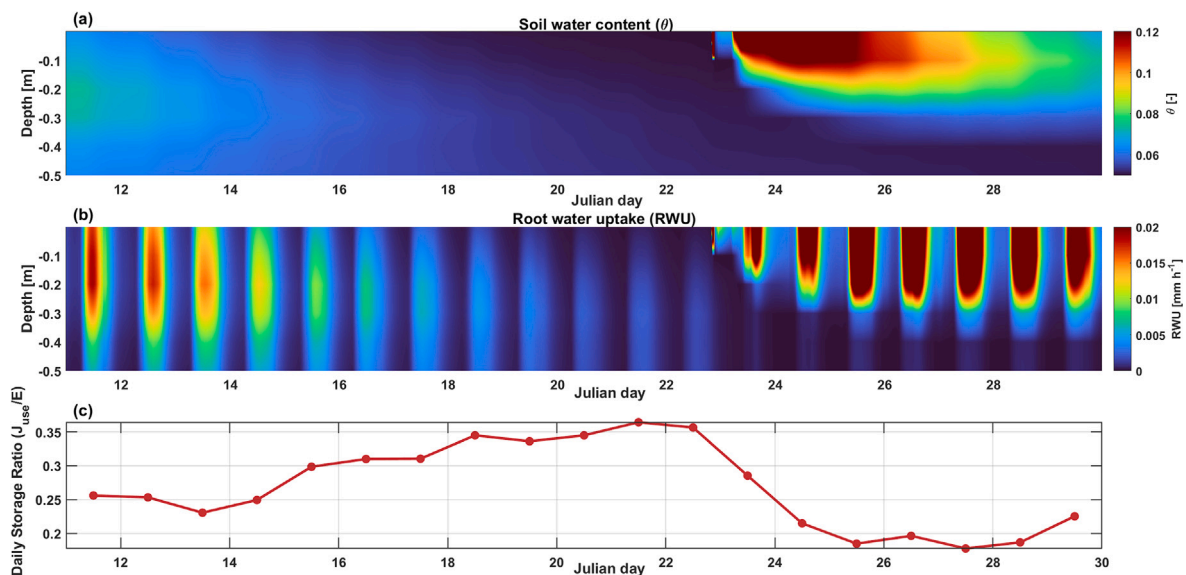


Fig. 10. Soil moisture, root water uptake, and daily use of internal water storage under varying soil moisture conditions. (a) Temporal evolution of soil volumetric water content (θ) in the upper 0.5 [m], illustrating a dry-down period followed by recovery. (b) Depth-time distribution of root water uptake (RWU) showing a downward shift in uptake under low soil moisture and rapid resumption of uptake once moisture availability increased. (c) Daily fraction of transpiration supplied by internal storage (J_{use}/E), which increased under low soil moisture and declined after rewetting.

storage to meet transpiration demand. Volumetric water content (θ) decreased steadily during days 11–23, marking the period of limited soil moisture availability. This dry-down period aligned with shifts in root uptake and increased reliance on stored water to support transpiration.

The temporal evolution of soil volumetric water content (Fig. 10 a) shows a distinct phase of progressive soil moisture depletion followed by recovery. From days 11–23, θ declined steadily due to evapotranspirative losses in the absence of compensatory rainfall, progressively reducing water availability for soil-root uptake. A sharp increase in θ around day 23 due to rainfall initiated the recovery phase. During days 24–30, soil moisture remained high and stable, indicating that the upper profile had rehydrated and moisture was no longer a limiting factor for root uptake.

The depth-time distribution of RWU (Fig. 10(b) shows a clear reduction in uptake from the upper soil layers as soil moisture declined. During the dry-down period, declining θ lowered effective uptake capacity because the soil-moisture-dependent stress reduction function ($f(\theta)$) suppressed potential uptake under limited soil water availability. Following the rainfall event on day 23, RWU increased sharply, demonstrating the rapid restoration of uptake with improved soil moisture and the close correspondence between soil water availability and root extraction.

The daily fraction of transpiration supplied by internal water storage increased steadily during the dry-down period, indicating that stored water played an increasingly important role as RWU decreased due to low soil water availability (Fig. 10 c). As surface θ declined from day 11 to 23, storage use increased from roughly 0.22 to 0.36, demonstrating a substantial shift in the relative contribution of stored water to daily fluxes, here quantified as J_{use} (the total daily volume of water withdrawn from internal storage). On average, storage contributed 37.1% more to transpiration during days classified as moisture-limited than during days with higher availability. Here, moisture-limited days were defined as $\theta < 0.09$, corresponding to the field capacity of the upper sand layer (Supporting Information Table S1), which is the horizon depicted in Fig. 10(a) and shows clear moisture fluctuations during the dry-down period. Following rainfall on day 23, storage use dropped sharply from about 0.36 to 0.17 during days 23–28, before rising around day 29 as soil moisture began to decline again. Statistical analysis confirmed a strong and consistent link between declining soil moisture and enhanced reliance on stored water: the relationship between daily θ and J_{use}/E was strongly negative (Spearman $\rho = -0.763$, $p < 0.001$; Pearson $r = -0.555$, $p = 0.014$), indicating that storage usage increased as soil moisture became limiting.

4. Discussion

Simulated and observed soil-plant hydraulic dynamics agreed closely across soil and environmental conditions examined here, supporting the model's ability to capture soil-plant hydraulic coupling with high accuracy (Fig. 6) at this site. Nevertheless, certain structural assumptions introduce limitations that may account for the remaining differences between modelled and observed values. The exclusion of temperature-dependent variations in water viscosity and plant hydraulic conductivity represents a notable simplification, as these properties are known to vary substantially with the seasonal temperature fluctuations experienced during the study period (Likhachev, 2003; Tyree and Zimmermann, 2013). Similarly, the assumption of a constant leaf area index throughout the simulations overlooks potential phenological shifts from summer to early winter, when canopy density and leaf water status are likely to vary (Rogers et al., 2021). Additional simplifications, such as fixed boundary layer conductance and idealized radiative forcing, may further constrain the model's ability to fully represent the coupled dynamics of atmospheric demand and plant water transport (Bunce, 1985).

The relatively high contribution of storage to total daily transpiration in our results, particularly during morning hours, reflects the potentially underrepresented role of root storage flux (Fig. 7). Previous experimental studies have primarily reported stem or shoot storage contributions

ranging from 2–50%, while typically neglecting the belowground component (Goldstein et al., 1998; Phillips et al., 2003). Accounting for root storage in our model naturally increases the total storage flux, providing a plausible explanation for the elevated morning values observed in the simulations. Our findings for this site and parameterization suggest that both root and shoot compartments contribute substantially to meeting early-day transpiration demand, which may be underestimated in approaches that consider only aboveground storage (Huang et al., 2017; Hartzell et al., 2017; Mirfenderesgi et al., 2016; Vogel et al., 2017).

Storage functions primarily as a temporal buffer, limiting the rate at which stress develops and reducing the cumulative hydraulic stress (Fig. 9). This buffering function becomes increasingly important as water potentials approach critical thresholds (Meinzer et al., 2004; Scholz et al., 2007, 2008). The consistent hierarchy across all dynamic metrics, including $d\psi/dt$, peak rates, and cumulative stress, indicates that shoot storage provides greater buffering, while root storage also plays a substantial role (Fig. 9). This indicates that the functional significance of internal storage depends not only on its total magnitude but also on its spatial distribution within the plant. Internal water withdrawal follows a clear temporal hierarchy, proceeding from distal to proximal organs (Scholz et al., 2008). Our results are consistent with this withdrawal hierarchy. As storage compartments are progressively removed, the phase lag between transpiration and RWU narrows systematically, confirming that each compartment contributes independently to the temporal decoupling between atmospheric demand and soil water supply. Shoot storage provides the dominant buffering, while root storage, proximal to the soil and hydraulically distinct from aboveground tissues (Huang et al., 2017; Scholz et al., 2007), provides supplementary buffering at a later phase of the diurnal cycle. Model results suggest that the effects of root and shoot storage are temporally complementary rather than functionally interchangeable, such that increasing aboveground capacitance alone cannot fully reproduce the timing and phase-offset effects associated with root storage (Chuang et al., 2006; Scholz et al., 2008). When root storage is removed in this framework, shoot storage flux responds substantially, increasing by 15%, and RWU increases by 7.5%, yet peak transpiration still declines by 7%, confirming that the compensation deficit is structural and spatial rather than volumetric. This spatial separation of storage functions is also reflected in nocturnal dynamics. The model partitions nocturnal root water uptake between transpiration, shoot storage refilling, and root storage refilling in proportions consistent with field observations at this site (Zeppel et al., 2010).

The consequences of this structural non-interchangeability of these compartments are most evident in the dynamics of hydraulic stress in the present model framework. Removal of shoot storage increased the maximum rate of stress development by 38% relative to the full storage case, whereas removal of root storage increased it by 13%. These independent effects indicate that each compartment buffers a distinct phase of the hydraulic stress trajectory (Meinzer et al., 2004; Hartzell et al., 2017; Vogel et al., 2017). Models that represent storage as a single lumped capacitance may reproduce aggregate fluxes while still misrepresenting the temporal evolution of plant water stress. The cumulative impact is substantial. Omitting shoot storage increases time-integrated hydraulic stress by 176%, while omitting root storage increases it by 129% at the xylem vulnerability threshold of -2.0 [MPa] over the mean diurnal cycle in the present framework. Consistent increases were also observed across additional ecologically grounded thresholds of -1.8 [MPa] and -1.9 [MPa], with effects expected to compound under prolonged drought. Since embolism formation depends on both the intensity and duration of xylem tension, these deviations in the predicted stress trajectory in the model may lead directly to biased estimates of hydraulic failure risk. These findings have important implications for plant hydraulic models, suggesting that approaches that neglect any storage compartments or treat capacitance as a minor correction factor may accurately predict instantaneous water potentials but will substantially underestimate both the rate at which plants approach critical thresholds and the cumulative stress experienced during prolonged exposure to low water potentials.

Soil moisture limitation plays a pivotal role in shaping how plants regulate internal water storage to sustain transpiration (Hartzell et al., 2017). During periods of low soil moisture or dry-down, plants increasingly rely on internal water storage to maintain stomatal function and support transpiration (Kumagai, 2001). The results observed in our simulations (Fig. 10) indicate that internal water storage acts as a buffer, decoupling transpiration from the corresponding root-uptake flow. This buffering mechanism provides critical hydraulic leeway, enabling the plant to maintain transpiration under drought stress in the simulated framework. This compensatory mechanism is demonstrated by the finding that stored water contributed 37.1% more to transpiration during moisture-limited days than during periods of higher availability in this framework. The increased dependence on internal reserves during dry-down periods is consistent with extensive empirical and modelling studies (Huang et al., 2017; Kumagai, 2001; Verbeeck et al., 2007). The alternation between RWU and storage release during the drying phase reflects the dynamic nature of water redistribution within the plant. This dynamic compensation has been shown to be vital for tree physiological functioning, delaying the onset of water stress and stomatal closure that would otherwise occur under restricted RWU in drying soil conditions (Hartzell et al., 2017).

The present model distinguishes the functional roles of root and shoot storage compartments across variable soil moisture and atmospheric conditions, showing that internal storage serves as a key hydraulic buffer that supports drought tolerance and stabilizes plant water transport. Root storage, proximally within the uptake domain, acts as a hydraulic reserve that sustains water delivery when soil-root fluxes lag behind transpiration (Fig. 8), thereby reducing the propagation delay of hydraulic stress. Shoot storage, positioned close to the transpiring leaves, provides the dominant temporal buffering, maintaining phase lag between transpiration and RWU (Huang et al., 2017; Hartzell et al., 2017). Omitting storage in any compartment leads to underestimation of transpiration and substantial overestimation of hydraulic stress, underscoring the importance of spatially distributed internal storage in regulating plant hydraulic function under the conditions examined here.

5. Conclusion

The daily integration of soil-plant fluxes reveals how plants maintain their hydraulic function, with internal storage acting as a capacitor that absorbs excess water during periods of low demand and releases it during periods of high demand. In the present model framework, this capacitive behaviour supports several critical physiological functions. First, it allows transpiration to respond rapidly to environmental conditions without being constrained by the slower dynamics of root water uptake. Second, it provides hydraulic buffering that prevents water potentials from approaching critical thresholds associated with cavitation and hydraulic failure. Third, it supports stomatal opening and gas exchange during periods when root water uptake alone would be insufficient to meet transpiration demand (Huang et al., 2017; Hartzell et al., 2017).

The quantified storage contributions and their distinct temporal dynamics offer site based insights relevant to improving soil-plant hydraulic models. Vegetation and land-surface schemes that represent plant capacitance as a single lumped parameter may not capture the functional differences revealed here, where the spatial location of storage along the hydraulic pathway appears to be as important as its magnitude within the present modelling framework. Models that ignore root storage entirely, dismissing it due to small flux contributions, risk underestimating its influence on sustaining transpiration and mitigating hydraulic stress. The incomplete compensation observed when either storage component is removed, despite substantial increases in the remaining storage fluxes and RWU, indicates that accurate prediction of plant hydraulic behaviour requires explicit representation of spatially distributed storage compartments within the present modelling framework. This finding is consistent with the temporal hierarchy of water

withdrawal established in the literature (Scholz et al., 2008) and is further supported by the compartment-specific partitioning of nocturnal root water uptake observed in this study (Huang et al., 2017; Scholz et al., 2008).

Despite these advances, several limitations remain. The model was applied at a single forest site, and the parameterization of storage properties relied on available data rather than direct measurements of tissue capacitance across plant organs. Furthermore, the model does not yet couple storage dynamics with structural acclimation, such as changes in root allocation or xylem anatomy under long-term drought. Future work should incorporate multispecies comparisons, direct measurements of root and shoot capacitance, and integration of storage with adaptive plant traits. Extending these analyses to ecosystem-scale models would also allow assessment of how storage-mediated hydraulic buffering influences surface fluxes, drought resilience, and water-energy feedbacks under a changing climate.

CRedit authorship contribution statement

Vinod Shankar Pathak: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Venkatraman Srinivasan:** Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at doi:10.1016/j.jhydrol.2026.135860.

Data availability

The datasets used in this study have been provided as supporting files for peer review. The code used to run the model will be made openly available upon publication at: <https://github.com/vinodspathak/vSPAC-1D.git>.

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